AI-BASED PERSONALIZED BOOK RECOMMENDATION ENGINE

PROJECT REPORT

Submitted by

SOJA VIJAY K

KMC24MCA-2032

to

the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree

of

Master of Computer Applications

Department Of Management Studies & Computer Applications

KMCT College of Engineering

Kallanthode, NITC P.O, Kozhikode-673601

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DECLARATION

I hereby declare that the project report “**AI-BASED PERSONALIZED BOOK RECOMMENDATION ENGINE** ”, submitted for partial fulfillment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Ms. Resmi SR**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Kallanthode SOJA VIJAY K

Date:

DEPARTMENT OF MANAGEMENT STUDIES & COMPUTER APPLICATIONS

KMCT COLLEGE OF ENGINEERING

Kallanthode, NITC P.O, Kozhikode-673601

CERTIFICATE

This is to certify that the report entitled **AI-BASED PERSONALIZED BOOK RECOMMENDATION ENGINE** submitted by **SOJA VIJAY K (KMC24MCA-2032)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications is a bonafide record of the project work carried out by her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Internal Supervisor	Project Coordinator
External Examiner	Head of the Department

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ABSTRACT

The AI-BASED PERSONALIZED BOOK RECOMMENDATION ENGINE is a web based system design to develop an AI powered book recommendation engine that helps users to discover books according to their reading preference. The system leverages machine learning techniques such as content-based filtering and collaborative filtering to analyze user behavior and book metadata. The application is built using Python for backend processing, with libraries like Scikit-learn while the front end is developed using Html/Css for a simple and interactive user interface. The core functionality includes user input for favorite books or ratings, real-time recommendation generation, and displaying similar books based on user preferences. AI plays a crucial role by applying algorithms that learn from user choices and suggest books with similar themes, genres, or patterns using similarity scores. In conclusion, this system offers a smart and efficient way to enhance the reading experience by providing personalized book suggestions through artificial intelligence.

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Chapter 1

INTRODUCTION

The **AI-Based Personalized Book Recommendation Engine** is a smart web-based system designed to help users discover books that align with their individual reading preferences. As digital reading platforms and online bookstores expand, readers often face difficulty navigating the vast volume of available content. This project addresses that challenge by introducing an AI-powered recommendation system that analyzes user behavior, reading patterns, and book metadata to generate personalized book suggestions.Using advanced machine learning techniques such as content-based filtering and collaborative filtering, the system identifies relationships between users and books to deliver accurate and meaningful recommendations. By incorporating real-time data processing and feedback mechanisms, the platform continually improves its recommendations, enhancing the user's reading experience. The interface is developed with HTML and CSS for a clean and interactive user experience, while the backend utilizes Python and machine learning libraries like Scikit-learn.In addition to providing convenience and personalization, the recommendation engine promotes the discovery of new titles, genres, and authors that users may not encounter otherwise. The system reflects a broader shift toward intelligent, data-driven applications that prioritize user engagement and adaptability.The application is designed with both users and administrators in mind. While users receive customized reading suggestions and can track their reading history and favorites, administrators are empowered to manage book metadata, monitor system feedback, and ensure content quality.Thus the project aims to create an intelligent, efficient, and user-centric platform that enhances the reading experience for book lovers of all kinds.

1.1 General Background

The growing popularity of online reading platforms, e-libraries, and digital bookstores has given users access to an overwhelming number of books. However, without effective guidance, this abundance can lead to confusion and indecision. Traditional recommendation methods—such as best-seller lists or manual browsing—often fail to consider the reader's personal tastes, resulting in a less satisfying reading experience.In response to this challenge, AI-powered recommendation systems have become essential tools for digital content platforms. These systems are capable of learning from user preferences and behavior to provide curated suggestions. By using data-driven approaches such as similarity scores, user profiling, and pattern recognition, the book recommendation engine transforms a passive browsing process into an intelligent, personalized experience.The demand for such systems highlights the importance of combining technology with human-centered design. Through personalization, users can explore content more efficiently, stay engaged, and expand their reading horizons, making AI-driven recommendations a core feature of modern reading platforms.

1.2 Objective

The objective of the AI-Based Personalized Book Recommendation Engine is to design and implement a smart, user-centric system that can generate customized book suggestions based on individual preferences, reading history, and user feedback. The system aims to enhance the book discovery process by leveraging artificial intelligence techniques, particularly content-based filtering and collaborative filtering, to analyze patterns in user behavior and book metadata. By creating dynamic user profiles and continuously updating them with real-time inputs such as ratings and reviews, the recommendation engine strives to offer increasingly accurate and relevant suggestions. Furthermore, the project seeks to build an intuitive web interface that enables users to easily interact with the system, provide feedback, and explore recommended titles effortlessly. Overall, the objective is to deliver a personalized and engaging reading experience by intelligently matching users with books that align with their unique interests and habits.

1.3 Scope

The AI-Based Personalized Book Recommendation Engine is designed to assist readers in discovering books that align with their individual preferences and reading patterns. The system allows users to register, create profiles, and update their favorite genres, authors, and ratings to receive personalized book suggestions. It manages user data securely and continuously adapts recommendations based on user interactions and feedback. Administrators can efficiently manage book records, including adding, editing, and deleting titles, as well as monitoring system performance and user activity.The system also provides functionalities such as book search, viewing reading history, rating books, giving feedback, and marking favorites, ensuring an engaging and interactive user experience. By combining algorithms like TF-IDF, Cosine Similarity, and Collaborative Filtering, the system delivers highly relevant and dynamic recommendations. Designed as a scalable and intelligent platform, it can accommodate an expanding user base and dataset, supporting future integration with advanced AI models and analytics. Overall, this project provides a flexible, efficient, and user-friendly solution for enhancing the way readers explore and enjoy books.

Chapter 2

LITERATURE SURVEY

The integration of artificial intelligence into personalized recommendation systems has significantly transformed how readers discover books that match their interests. According to Singh et al. (2023) [1], AI-driven book recommendation engines that analyze user preferences, reading history, and behavior enable highly personalized and context-aware suggestions. These systems aggregate data from various user interactions to understand individual tastes, thereby enhancing user satisfaction and engagement. Patel and Mehta (2024) [2] highlight that user-friendly interfaces in book recommender platforms significantly improve user interaction and usability by offering customizable reading dashboards, genre-based filters, and smart suggestions. Their findings suggest that intuitive design and personalized content presentation play a crucial role in improving the effectiveness of recommendation systems.

In their book AI in Personalized Content Delivery, Varma and Iyer (2023) [3] provide comprehensive insights into hybrid recommendation techniques, combining content-based filtering (based on book genres, author, and metadata) and collaborative filtering (based on user similarity and preferences). This approach is shown to improve both the accuracy and diversity of book recommendations. Similarly, Rajan (2024) [4] explores how real-time AI-based personalization helps readers discover relevant content quickly by adapting to their most recent reading activities. Adaptive recommenders, which update suggestions based on evolving preferences, are increasingly recognized for keeping users engaged in digital reading platforms.

The website BookTechReview.org (2025) [5] explores the practical applications of AI-powered book recommendation engines, stating that predictive modeling and machine learning algorithms reduce user effort by automatically curating book lists tailored to individual interests. Nair and Krishnan (2023) [6] emphasize that these systems offer time-saving benefits and increase recommendation accuracy by learning from user behavior patterns. According to Menon and Pillai (2022) [7], integrating hybrid AI models into recommendation engines improves user retention and system efficiency by offering more relevant and diverse book choices while minimizing repetitive suggestions.Further, Thomas et al. (2021) [8] stress the importance of real-time personalization, which ensures that recommendation engines remain responsive and adaptive as new books are added or as user interests shift. Ramesh and Gupta (2020) [9] highlight that personalized recommendations improve content discovery and user satisfaction by reducing search friction. Meanwhile, Bhatia and Rao (2021) [10] discuss how automation streamlines metadata processing and enhances recommendation quality, leading to better content matching and engagement.

Additionally, modern recommendation engines are increasingly supported by cloud and mobile technologies. Joseph and Varma (2023) [11] state that cloud-based platforms enable seamless synchronization of user profiles and reading preferences across devices, facilitating uninterrupted access to personalized book recommendations. Iyer and Sreedhar (2022) [12] explore the growing adoption of mobile-first AI recommenders, allowing users to receive timely suggestions on the go. Das and Nambiar (2021) [13] underline the value of location-aware recommendations and user segmentation, while Fernandes and Kumar (2020) [14] highlight the benefits of offering customizable recommendation settings to suit different reader types, including casual readers, students, and researchers. Collectively, these studies confirm that AI-based personalized book recommendation engines are playing a pivotal role in improving reader satisfaction, recommendation accuracy, and platform efficiency in digital content ecosystems.

Chapter 3

SYSTEM ANALYSIS

3.1 Existing System

In today's digital world, users have access to millions of books online, but most traditional recommendation systems fail to give suggestions that truly match a person's unique interests. These older systems usually suggest bestsellers or general popular books without understanding the individual reader's taste. This often leads to user dissatisfaction because they have to spend time searching for books they actually like. To solve this problem, the proposed AI-based personalized book recommendation system uses smart techniques to learn what each user likes and give suggestions that are truly personalized. The system first collects information about the user's preferences—such as the genres they read, the authors they follow, or how they rated other books. At the same time, it gathers book metadata, which includes keywords, genre, author, and summary of each book.

The system then uses a method called TF-IDF (Term Frequency-Inverse Document Frequency) to understand the important words in each book's description.After that, a technique called cosine similarity is used to compare the user's interests with all the books in the database. It checks which books are most similar to the user's profile based on the words and topics they care about. Once the system finds the books with the highest similarity scores, it selects the top ones and shows them as recommendations to the user. This whole process happens automatically and very quickly, helping the user discover books they are likely to enjoy. Over time, the system learns from the user's reading behavior and continues to improve its recommendations.It saves time, improves satisfaction, and helps readers explore books that match their exact interests—making the platform smarter and more user-friendly.

3.2 Proposed System

The proposed system aims to solve a common challenge faced by readers today — the difficulty of finding the right book among thousands of available options. Unlike traditional systems that recommend only trending or popular titles, this project uses Artificial Intelligence (AI) to deliver personalized book suggestions based on each user's unique preferences and reading behavior. The system intelligently analyzes user interests and patterns to provide recommendations that are accurate, meaningful, and continuously improving.The system begins by collecting and storing user data such as favorite genres, preferred authors, reading history, and book ratings. This information is used to create a personalized user profile that reflects the individual's literary taste. As users interact with the platform — by rating or liking books — their profiles are automatically refined, allowing the system to adapt dynamically to changing interests and habits. This adaptive learning process ensures that the more a user engages with the system, the more relevant the recommendations become. The system relies on a structured dataset containing information like book titles, authors, genres, publication details, summaries, keywords, and user ratings. To ensure accuracy and efficiency, the data undergoes preprocessing steps such as cleaning, tokenization, stop-word removal, and normalization. These steps make the dataset suitable for machine learning tasks like vectorization and similarity analysis, which are essential for accurate recommendation generation. To understand and represent books mathematically, the system uses the TF-IDF (Term Frequency–Inverse Document Frequency) algorithm. TF-IDF identifies the most important words in each book's description, transforming textual information into numerical vectors that capture each book's main characteristics. The system then uses the Cosine Similarity algorithm to compare the user's profile vector with book vectors, measuring how closely a book aligns with a user's interests. Based on these similarity scores, books are ranked and the most relevant titles are presented to the user. In addition to content-based filtering, the system implements Collaborative Filtering to enhance the diversity and accuracy of results. Collaborative filtering identifies patterns among users with similar reading habits and recommends books enjoyed by like-minded readers. By combining Content-Based Filtering (TF-IDF and cosine similarity) with Collaborative Filtering, the system forms a Hybrid Recommendation Model that utilizes the advantages of both approaches to deliver balanced and precise recommendations. The system supports two main user types — end users (readers) and administrators. End users can register, log in, explore book recommendations, rate titles, and provide feedback, which helps improve the recommendation accuracy. They can also search for books, mark favorites, and view reading history. The administrator manages the system from the backend, ensuring smooth performance by adding or updating book records, monitoring user data, and maintaining dataset quality. The admin panel is designed for easy control over system operations while users enjoy a seamless reading experience. Overall, this AI-Based Personalized Book Recommendation Engine offers an intelligent and adaptive solution for discovering books. By integrating algorithms such as TF-IDF, Cosine Similarity, and Collaborative Filtering, the system effectively learns from user interactions to deliver accurate and engaging recommendations. This approach not only saves time but also enhances the user's reading experience, demonstrating how AI can personalize and enrich digital platforms in an evolving technological world.

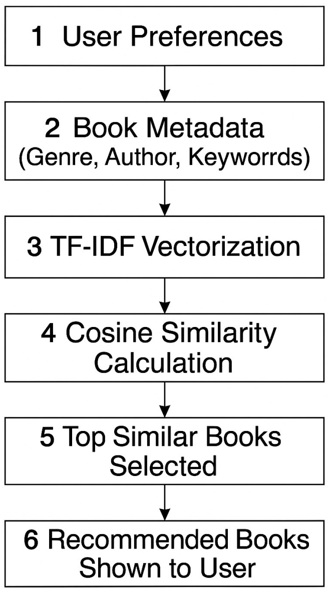


Figure 3.1: Book Recommendation System Flowchart

3.3 Module Description

3.3.1 User Profile

The User Profile Module is responsible for managing individual user data, capturing their preferences, and maintaining a history of interactions. It stores key information such as favorite genres, previously rated books, and personalized settings. By continuously updating these preferences based on user behavior, the system builds a dynamic user profile that serves as the foundation for generating relevant recommendations.This module ensures that the system captures the unique tastes and interests of users, enabling the recommendation engine to tailor suggestions more accurately.

3.3.2 Recommendation Engine

The Recommendation Engine is the core AI-driven component that analyzes the user profile and book metadata to suggest titles tailored to each user's reading habits. This module uses content-based filtering to compare book features and collaborative filtering to analyze user patterns across the system. By computing similarity scores and leveraging machine learning algorithms, it delivers accurate and engaging recommendations in real time.This module ensures that users receive timely and meaningful book recommendations aligned with their individual tastes

3.3.3 Rating And Feedback Module

The Rating And Feedback Module allows users to provide feedback on the recommended books, either through ratings or textual input. This feedback is instrumental in refining the recommendation system, helping the AI learn more accurately from user interactions. Over time, the system adapts and improves its suggestions, creating a more personalized and satisfying reading experience for each user.This module ensures that the recommendation system evolves and improves over time by learning directly from user interactions.

3.4 Feasibility Study

The feasibility study helps us understand whether the AI-Based Personalized Book Recommendation Engine can be built and used successfully. This system aims to suggest books to users based on their personal reading preferences, using artificial intelligence. To make sure the system can be implemented effectively, we examine its operational, technical, and economic aspects. Each part of this study ensures that the system is realistic, useful, and beneficial in a real-world setting.

3.4.1 Operational Feasibility

From an operational point of view, the system is highly feasible. Users will be able to easily sign up, input their interests, and receive book suggestions tailored just for them. The interface will be designed to be simple and smooth, so that even people who are not tech-savvy can use it comfortably. The system will continuously learn from user behavior, such as the books they rate highly or the ones they search for, and improve the recommendations over time. Administrators or librarians can also use the system to track popular genres or frequently recommended books. This can help them manage book inventories or understand what readers are currently interested in. Since the system is designed to work in the background and only requires users to interact when needed, it will not disturb the regular operations of a website or app it is added to.Additionally, users can access their recommendations on any device like mobile phones, tablets, or laptops, making it flexible and convenient. With proper testing and training, even first-time users can quickly adapt to the platform.

3.4.2 Technical Feasibility

The technical side of the project is practical and realistic. The recommendation engine will use machine learning techniques such as TF-IDF (Term Frequency-Inverse Document Frequency) to process the text data of books (like titles, genres, and keywords). It will then use cosine similarity to compare user interests with book data and find the most suitable books for each user. All of this can be implemented using programming languages like Python and its popular libraries such as Scikit-learn. For the web interface, we can use frameworks like Python for the backend, and HTML, CSS, and JavaScript for the frontend. These tools are open source, widely used, and well-documented, which makes the development process easier.Moreover, the system can be hosted on a basic server or cloud platform such as Heroku, AWS, or Firebase. Since it's a lightweight application in its early stages, it won't need heavy computational power, making it technically manageable for students, startups, or small organizations.

3.4.3 Economic Feasibility

Economically, the system is very affordable to build and run. Since it uses free and open-source tools, there is no need to purchase expensive software or licenses. The developers only need a computer with a decent configuration and an internet connection. Deployment costs can also be minimized by using low-cost hosting services or free tiers on platforms like GitHub Pages or Firebase. The biggest cost is time and effort for development and testing, but no major financial investment is required. This makes it suitable for academic projects, libraries, small bookshops, or individual developers who want to create personalized experiences for readers. In the long term, the system can reduce the time people spend looking for books, improve user satisfaction, and help businesses or libraries provide better services, which makes it economically beneficial.

3.5 System Environment

3.5.1 Developer Requirement

3.5.1.1 Hardware Requirement

- Processor: Intel Core i5 or above
- RAM: 8 GB RAM or above
- Storage: Atleast 500 GB of hard disk space
- Graphics: Basic graphics support for running development environment

3.5.1.2 Software Requirement

- Operating System: Windows 10 or Ubuntu
- Frontend: HTML/CSS,Javascript

- Backend: PYTHON
- Database: MySQL
- ML Tools:Python,Scikit-learn
- IDE: VS Code
- Browser: Chrome or Firefox

3.5.2 User Requirement

- Device: Any smartphone,tablet or computer
- Internet Connection: Stable internet for loading recommendation and submitting preferences
- Browser: Any modern web browser such as Chrome or Firefox

3.6 Actors and Their Roles

3.6.1 User

- Registers and logs in to the system
- Enters preferences like favorite genres, authors, etc.
- Views personalized book recommendations
- Searches for books by keyword, genre, or author
- Rate and feedback books after reading
- Marks books as favorites
- Updates their profile and reading preferences
- Tracks their reading history

3.6.2 System Administrator

- Manages user accounts (add, delete, or update users)
- Adds, edits, or removes books in the system

- Monitors system performance and ensures it runs smoothly
- Views reports on user activity and book usage
- View feedback submitted by users

Chapter 4

SYSTEM DESIGN

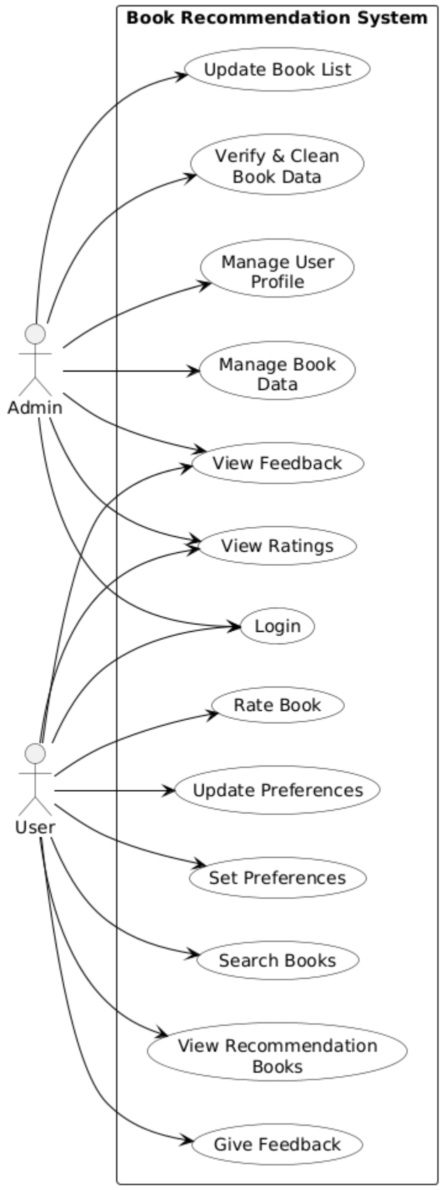
4.1 Methodology

This project adopts the Agile methodology for system development. Agile emphasizes iterative and incremental development, allowing continuous improvement through frequent collaboration among developers, users, and stakeholders. This approach enables the system to adapt to changing user needs and technical challenges throughout the development lifecycle.By dividing the development into small, manageable tasks called sprints, the team can collect early feedback from users and incorporate it into subsequent iterations. This ensures the recommendation engine evolves in a way that better reflects user preferences, reading behavior, and expectations. Agile encourages continuous testing and refinement of the algorithms used for book recommendation, such as TF-IDF and cosine similarity, ensuring improved accuracy over time. Frequent communication between team members and regular reviews of progress help in identifying potential improvements quickly. This is especially important for AI-driven systems where user interaction data and system feedback loops play a crucial role in enhancing personalization.Agile also supports flexible planning, making it easier to update features like preference settings, recommendation logic, or user interface components based on ongoing user feedback. As a result, the project remains user-centered, efficient, and aligned with the core objective — helping users discover books that truly match their interests.This methodology is particularly effective for dynamic applications like an

AI-based book recommendation system, where user patterns and preferences change over time, and the system must continuously adapt to stay relevant and useful.

4.2 UML Diagrams

4.2.1 Use Case Diagram



4.2.2 Activity Diagram

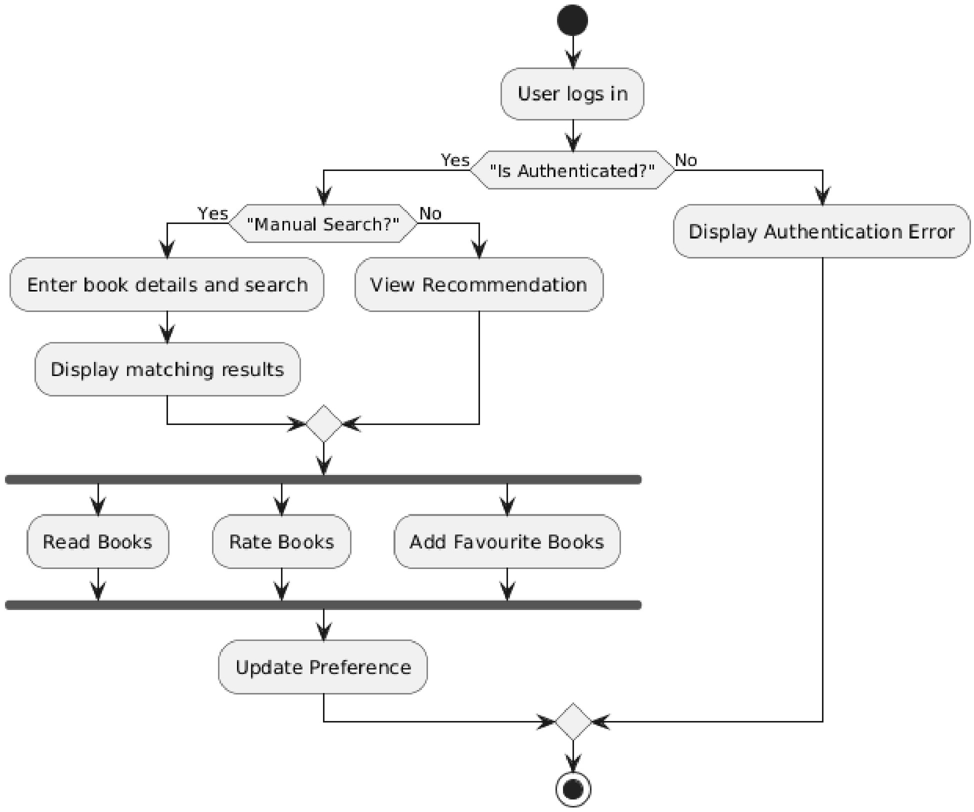


Figure 4.1: Activity Diagram

4.2.3 Class Diagram

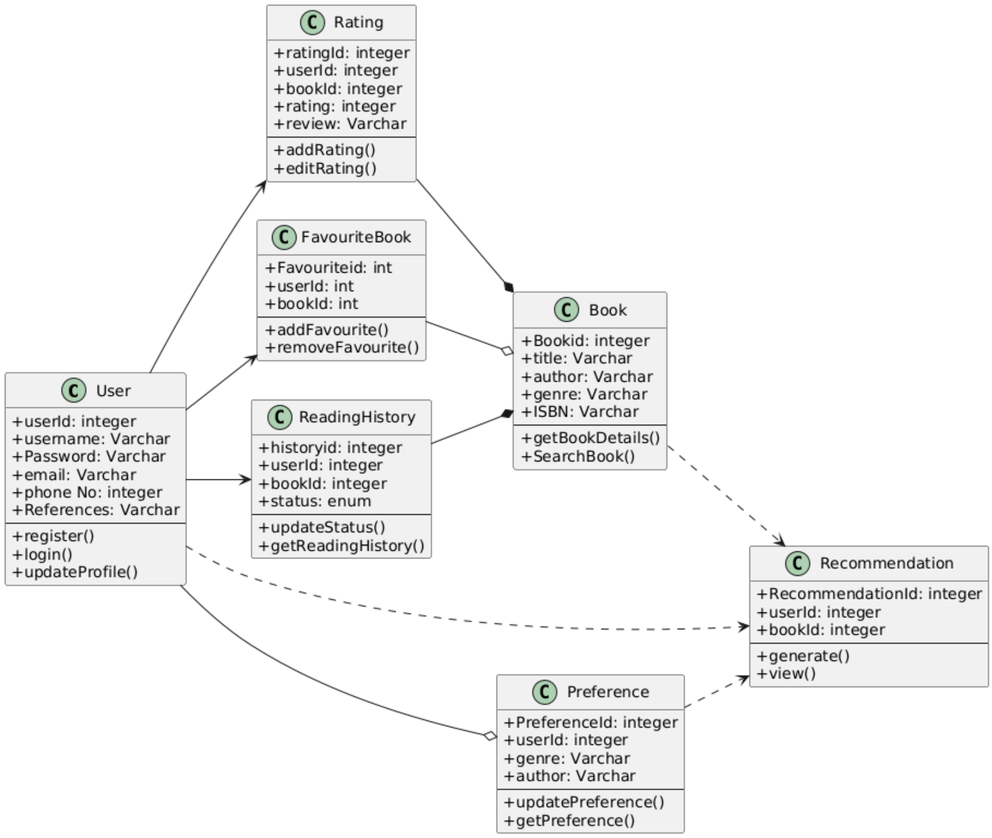


Figure 4.2: Class Diagram

4.3 User Story

User story ID	As a (user)	I want to...	So that I can...
1	User	To login with user name and password	To access my account and personalized book features
2	User	To register by entering my personal details and reading preference	To generate a personalized profile,so can recieve tailored book suggestions
3	User	To set and update my favorite genres and author	To help the recommendation engine suggest books I enjoy
4	Admin	To manage book data and view user profiles	To ensure books, user information are accurate,update generate relevent recommendations
5	User	To search for a book or enter one I already like	To discover new titles based on my interest
6	User	To view recommended books with titles,author,and summary details	To decide with suggested book to read
7	User	To read and rate book and to add into favorite	To improve future recommendation and to view favorite book
8	User	To give feedback about the recommendation	To share how useful the suggestions user and improve system relevence
9	Admin	To review user ratings and feedback trends	To understand satisfaction and adjust content accordingly
10	User	To add readed books,into reading history and view favorites	To keep track of books that already readed, to refine future history based on past activity

Table 4.1: User Story

4.4 Product Backlog

User Story ID	Priority (Low, High, Medium)	Size	Sprint	Status (Planned, Progressed, Completed)	Release Date	Release Goal
1	Medium	6	1	Completed	06-08-2025	Users can log in to access personalized features.
2	Medium	6		Completed	11-08-2025	Users can register to generate user profile.
3	Low	4		Completed	18-08-2025	Users can set and update their favorite authors/genres.
4	Medium	6	2	Completed	25-08-2025	Admin can manage book data and view user profiles.
5	High	10		Completed	01-09-2025	Users can search books by title, genre, or author.
6	High	10	3	Completed	10-09-2025	Users can view recommended books with summaries.
7	Medium	6		Completed	15-09-2025	Users can read, add to favorites, and rate books they've read.
8	Low	4		Completed	22-09-2025	Users can give feedback on recommendations.
9	Medium	6	4	Completed	29-09-2025	Admin can review user feedback and adjust recommendations.
10	High	10		Completed	08-10-2025	Users can view favorite books and reading history.

Table 4.2: Product Backlog

4.5 Project Plan

User Story ID	Task Name	Start Date	End Date	Days
1	Login page	04/08/25	06/08/2025	3
2	Registration page	08/08/25	11/08/2025	4
3	Manage favorite genre and author	12/08/25	18/08/2025	7
4	Manage book data and view user profile	19/08/25	25/08/2025	7
5	Discover new books by searching or already liked one	26/08/25	10/09/2025	8
6	View recommended books according to suggestions	26/08/25	10/09/2025	8
7	Read ,rate and books into favorites	11/09/25	15/09/2025	5
8	Feedback submission for already readed books	16/09/25	22/09/2025	7
9	View ratings and feedback given by user	23/09/25	29/09/2025	7
10	View favorites and reading history	03/10/25	08/10/2025	6

Table 4.3: Project Plan

4.6 Database Design

4.6.1 User Table

The ”User Table” stores essential user information, including login credentials, personal details, and contact data. It also maintains user preferences such as favorite genres and authors to support personalized recommendations. The “role” field distinguishes between users and administrators. Constraints like primary key, uniqueness, and not-null ensure data integrity and security within the system..

No	Name	Type	Constraints	Description
1	user_id	Integer	Primary key	Unique identifier for each user
2	username	Varchar	Not null	Username used for login
3	password	Varchar	Not null	Encrypted password for authentication
4	email	Varchar	Unique, Not null	User’s email address
5	phone number	Varchar	Unique, Not null	User’s contact phone number
6	dob	Date	Not Null	User’s dob
7	gender	Varchar	Not Null	User’s gender
8	genre	Varchar	Not Null	User preferred genre
9	author	Varchar	Not Null	User preferred author
10	role	Varchar	Not Null	Role that wheather user or admin

4.6.1 User Table

4.6.2 Book Table

The ”Book Table” stores detailed information about each book, including its title, author, genre, ISBN, and content. It ensures accurate cataloging through constraints like primary key and uniqueness. The table also includes a short description field to support content-based filtering and improve personalized book recommendations.

No	Name	Type	Constraints	Description
1	book_id	Integer	Primary key	Unique ID for each book
2	title	Varchar	Not null	Name of the book
3	author	Varchar	Not null	Author who wrote the book
4	genre	Varchar	Not null	Category of the book
5	ISBN	Varchar	Unique, Not null	International Standard Book Number
6	description	Varchar	Not null	Short summary of the book
7	content	Long text	Not null	Content of the book

4.6.2 Book Table

4.6.3 Favorite Book Table

The ”Favorite Book Table” records the books that users have marked as favorites. It establishes a relationship between users and books through foreign keys, ensuring data consistency. Each entry has a unique identifier, allowing efficient tracking and retrieval of a user’s favorite books for personalized recommendations..

No	Name	Type	Constraints	Description
1	fav_id	Integer	Primary key	Unique identifier for the favorite entry
2	user_id	Integer	Foreign key	References the user who marked a book as favorite
3	book_id	Integer	Foreign key	References the book that is favorited

4.6.3 Favorite Book Table

4.6.4 Rating Table

The ”Rating Table” stores user feedback as numerical ratings for books, linking each rating to the respective user and book through foreign keys. It includes constraints to ensure ratings remain within a valid range (1–5) and timestamps to record when each rating is created or updated, supporting accurate analysis of user preferences.

No	**Name**	**Type**	**Constraints**	**Description**
1	rating_id	Integer	Primary key	Unique ID for each rating entry
2	user_id	Integer	Foreign key	Refers to the user who gave the rating
3	book_id	Integer	Foreign key	Refers to the book that is rated
4	rating	Integer	Check (rating between 1 and 5)	Actual rating given by the user (1 to 5)
5	created at	Timestamp	Not null	Book rating created time
6	updated at	Timestamp	Not null	Book rating updated time

4.6.4 Rating Table

4.6.5 Feedback Table

The ”Feedback Table” records user comments or opinions about books, linking each feedback entry to the respective user and book through foreign keys. It ensures every feedback message is stored with a unique ID and includes timestamps for creation and updates, enabling effective tracking of user interactions and book evaluations.

No	Name	Type	Constraints	Description
1	feedback_id	Integer	Primary key	Unique ID for each feedback entry
2	user_id	Integer	Foreign key	References the user who gave the feedback
3	book_id	Integer	Foreign key	References the book the feedback is about
4	feedback	Varchar	Not null	Actual feedback message from the user
5	created at	Timestamp	Not null	Book feedback created time
6	updated at	Timestamp	Not null	Book feedback updated time

4.6.5 Feedback Table

4.6.6 Reading History Table

The ”Reading History Table” keeps track of books read by users, linking each record to the corresponding user and book through foreign keys. It includes a unique identifier and a status field that indicates the reading progress, helping the system analyze user activity and generate better personalized recommendations.

No	Name	Type	Constraints	Description
1	history_id	Integer	Primary key	Unique ID for reading history record
2	user_id	Integer	Foreign key	References the user who read the book
3	book_id	Integer	Foreign key	References the book that was read
4	status	Enum	Not null	Reading status of the book

4.6.7 Reading History Table

4.6.7 Reading Progress Table

The ”Reading Progress Table” records each user’s current progress while reading a book. It links users and books through foreign keys and stores details such as the last read page, paused word, and sentence for accurate progress tracking. Timestamps for creation and updates ensure data consistency and allow users to resume reading seamlessly.

No	Name	Type	Constraints	Description
1	progress_id	Integer	Primary key	Unique ID for reading progress record
2	user_id	Integer	Not null	References the user who reads the book
3	book_id		Foreign key	References the book that was reading
4	page_number	Integer	Not null	Number of the page where the progress is recording
5	paused_word	Varchar	Not null	Word that is paused where the progress is recording
6	paused_senetence	Text	Not null	sentence which the word is paused for progress recording
7	created at	Timestamp	Not null	Book feedback created time
8	updated at	Timestamp	Not null	Book feedback updated time

4.6.7 Reading Progress Table

4.7 User Interface Design

4.7.1 Login Page

The Book Bot login interface presents a clean and modern design, divided into two sections. On the left, a vibrant blue panel welcomes users with the message “Welcome to Book Bot,” branding it as a personal assistant for discovering new reads. An open book icon reinforces the reading theme. On the right, users are prompted to log in by entering their username and password. The interface also provides helpful links for password recovery (“Forgot password?”) and new user registration (“Register”), all styled for a smooth and user-friendly experience.

1.png

Figure 4.3: Login Page

4.7.2 Forgot Password Page

The ”Forgot Password” screen shown in the reference image is designed to assist users in securely resetting their account password. At the top of the screen, users are prompted to enter their registered email address into a clearly labeled input field. After entering the email, they can click on the ”Send Verification Link” button, which initiates the process of sending a secure verification link to the provided email. Once the email is verified, it is displayed in a masked format (e.g., ‘ex****@email.com‘) to ensure privacy. Below this, users are given fields to enter a new password and confirm it to avoid typing errors. The ”Change Password” button allows users to submit their new password securely. At the bottom of the screen, there is a link labeled ”Back to Login” for users who wish to return to the login page. The overall layout is simple and modern, using a white card-style panel centered on a light grey background, offering a clean and user-friendly interface.

2.png

Figure 4.4: Forgot Password

4.7.3 Create New Account

The login page of the AI-based Book Recommendation System, titled "Book Bot," is designed with a clean and user-friendly interface to welcome users into a personalized reading experience. The layout is split into two sections: the left side features a vibrant blue background with a welcoming message, "Welcome to Book Bot," accompanied by a brief tagline, "Your personal assistant for discovering new reads," which highlights the platform's purpose. On the right side, users are prompted to log in using their username and password through clearly labeled input fields. A "Forgot password?" link provides easy password recovery, while the "Register" option at the bottom invites new users to join the platform. The design ensures a smooth and intuitive entry point for users to access AI-powered book suggestions tailored to their reading preferences.

3.png

Figure 4.5: Create New Account

4.7.4 Author Selection

This page allows users to personalize their reading experience by selecting their favorite authors. Titled “Choose authors you like!”, the interface encourages users to pick preferred authors to receive more accurate and personalized book recommendations from the Book Bot. A search bar at the top helps users quickly find specific authors, while the main section displays a grid of author cards, each accompanied by a checkbox and an avatar for easy identification. Users can select multiple authors from the list, and a prominent “Save” button at the bottom ensures that their preferences are securely stored. This interactive layout plays a crucial role in tailoring AI-driven recommendations based on the user's literary interests.

4.png

Figure 4.6: Author Selection

4.7.5 Genre Selection

This page allows users to personalize their reading experience by selecting their favorite authors. Titled "Choose authors you like!", the interface encourages users to pick preferred authors to receive more accurate and personalized book recommendations from the Book Bot. A search bar at the top helps users quickly find specific authors, while the main section displays a grid of author cards, each accompanied by a checkbox and an avatar for easy identification. Users can select multiple authors from the list, and a prominent "Save" button at the bottom ensures that their preferences are securely stored. This interactive layout plays a crucial role in tailoring AI-driven recommendations based on the user's literary interests.

6.png

Figure 4.7: Genre Selection

4.7.6 Home Page

The Home Dashboard of the Book Bot interface offers a central hub for user navigation and personalized reading engagement. On the left, a vertical sidebar menu provides quick access to core sections including: Home,Recommended For You, Explore By Category, Recently Rated, Favorites, and Continue Reading. Each menu item is clearly marked with a relevant icon for intuitive access.The main panel greets the user with a welcoming message: ”Welcome! What would you like to read today?” directly above a search bar, allowing users to find books, authors, or genres swiftly. Below the search area, six prominent tiles present the main content categories. These include Recommended For You,Explore By Category, Recently Rated, Favorites, and Continue Reading, each paired with a visually distinct icon.The layout is clean and visually segmented, offering users a user-friendly experience for accessing personalized reading options and managing their book interactions efficiently. The top-right section features quick access links for Profile and Logout, maintaining accessibility and session control.

7.png

Figure 4.8: Home Page

4.7.7 Recommended For You

The ”Recommended for You” section in the Book Bot interface presents users with personalized book suggestions based on their preferences and reading history. Upon selecting this option from the sidebar, users are welcomed with a consistent header and a search bar to find specific books, authors, or genres.Below the header, the page dynamically displays recommended books in a visually organized card format. Each card includes a book cover image, title, author name, and a star rating system, allowing users to quickly assess popularity and quality. Two primary action buttons are provided: “Add to Favorites”, marked with a heart icon, and “Read Now”, highlighted in blue to encourage immediate engagement. This layout simplifies user decision-making while promoting interaction with AI-curated content that aligns with the user’s unique reading profile.

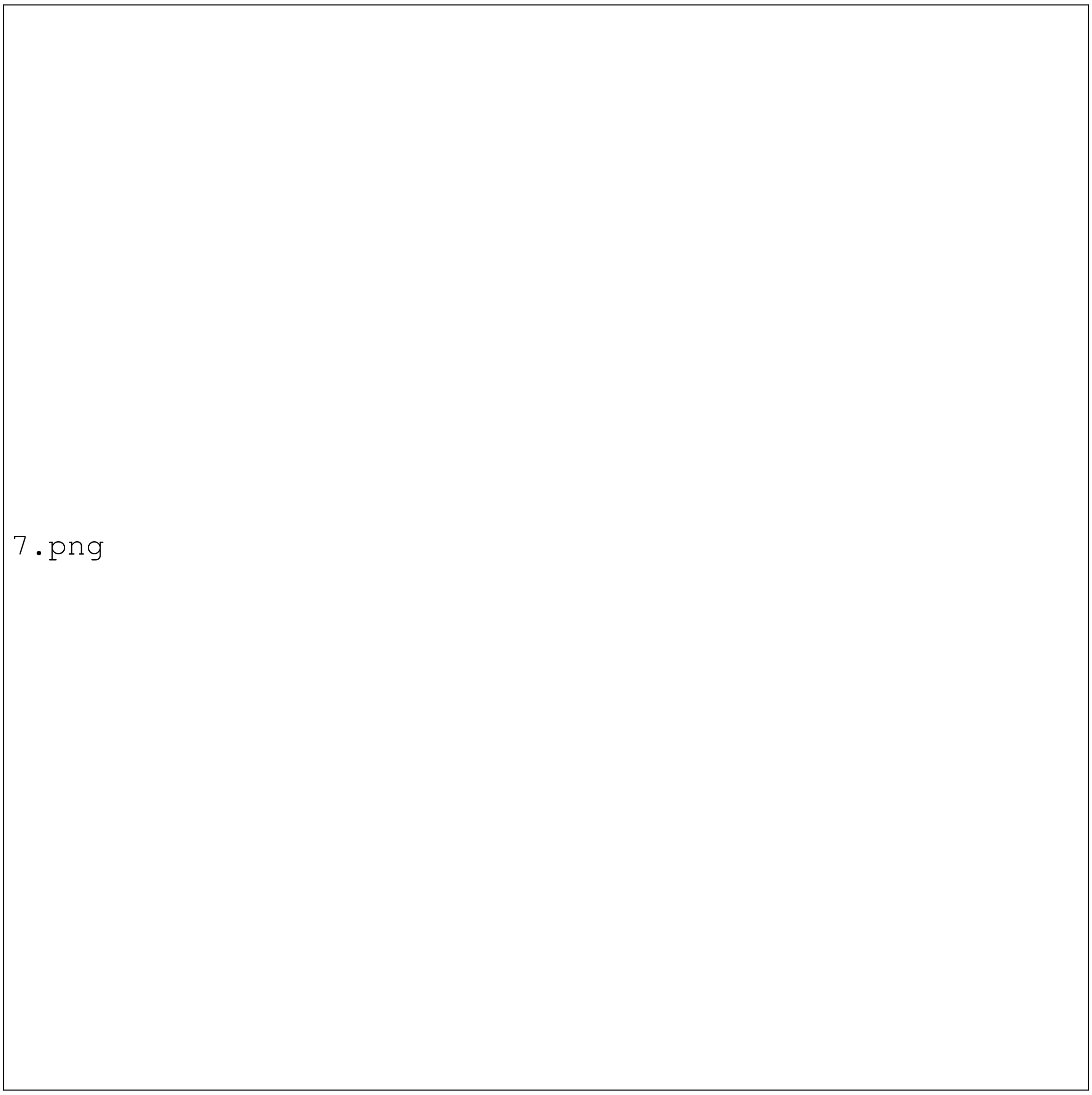


Figure 4.9: Recommended For You

4.7.8 Personalized Book Overview

The Home section of the Book Bot interface acts as the user's personalized dashboard, offering curated book recommendations. Under the "Recommended for You" header, the interface showcases a single featured book in a clean, horizontal layout. The display includes a book cover image, followed by book details such as the title (Book 1), author name, genre, and a star rating presented with visual stars.A brief description summarizes the plot and introduces the main characters to engage the reader. Users are provided with three intuitive action buttons: "Add to favorites", "Rate book", and "Read", enabling smooth interaction with the recommended content. This layout emphasizes clarity, reader engagement, and seamless access to essential book actions—all tailored for a personalized experience.

8.png

Figure 4.10: Personalized Book Overview

4.7.9 Explore by Books

The Explore by Books section of the Book Bot interface allows users to browse various titles based on their preferences. Displayed in a card layout, each book listing includes a cover image, book name, author, genre, star rating, and a brief description. For instance, the first book has a star rating and presents a short summary giving a quick insight into the plot and main characters. The second book, rated stars, features a different summary providing a glimpse into its unique storyline.Each book card includes action buttons such as “Add to Favorites” and “Read”, making interaction seamless. This layout promotes quick discovery while offering essential details that help users select books of interest efficiently.

9.png

Figure 4.11: Explore by Books

4.7.10 Dashboard Interface

The Book Bot dashboard interface offers a streamlined and user-friendly layout for discovering and exploring books. The left sidebar includes navigational links such as Home, Recommended for You, Explore by Category (Books, Authors, Genres), Recently Rated, Favorites, and Continue Reading. At the top, users can access their Profile or Logout. The main section greets users with a welcome message and a search bar for finding content. Below, under ”Explore by Books,” a featured book titled “Book 1” is displayed with a brief description and a prominent “Read Now” button, making it easy to start reading instantly.

10.png

Figure 4.12: Dashboard Interface

4.7.11 Story Reading Interface

The Book Bot story reading interface provides a focused environment for reading and engaging with book content. At the top, users can navigate pages (e.g., Page 1 of 250), perform searches, and adjust font size. The central section features the About Story heading with the story content displayed in a large, readable text box. Below the text, users have the option to pause or continue reading with dedicated buttons. On the right side, a star-based rating system allows readers to provide feedback, accompanied by a comment box for sharing thoughts and a green Save button to submit feedback.

11.png

Figure 4.13: Story Reading Interface

4.7.12 Browse By Authors

The Book Bot dashboard interface offers a streamlined and user-friendly layout for discovering and exploring books. The left sidebar includes navigational links such as Home, Recommended for You, Explore by Category (Books, Authors, Genres), Recently Rated, Favorites, and Continue Reading. At the top, users can access their Profile or Logout. The main section greets users with a welcome message and a search bar for finding content. Below, under "Explore by Books," a featured book titled "Book 1" is displayed with a brief description and a prominent "Read Now" button, making it easy to start reading instantly.

12.png

Figure 4.14: Browse By Authors

4.7.13 Browse by Genre

The Genres section within the “Explore by Category” menu in the Book Bot interface allows users to browse books based on thematic preferences. Under the header “Explore by Genres,” users are presented with a clean, grid-style layout showcasing selectable genre options such as Genre 1 through Genre 6. Each genre is paired with a checkbox for easy multi-selection.A prominent search bar at the top enables users to filter results by entering keywords related to books, authors, or genres. Once selections are made, clicking the “Read” button at the bottom right takes users to content matching their chosen genres. This layout ensures an intuitive and tailored browsing experience focused on user interests.

13.png

Figure 4.15: Browse by Genre

4.7.14 Recently Rated

The Recently Rated section in the Book Bot interface provides users with a clear and organized view of the books they have rated. Located in the left-side navigation, this section appears under the title “Recently rated” and displays a vertical list of previously reviewed books. Each entry includes a book cover image, the book name, a star rating , and a feedback text box for additional comments.A search bar at the top allows users to filter through books by title, author, or genre. This section serves as a personalized archive of user engagement, encouraging reflection and providing a space to share thoughts or edit feedback on past reads.

14.png

Figure 4.16: Recently Rated

4.7.15 Favorites

The Favorites section in the Book Bot interface displays a curated list of books that the user has marked as favorites. Under the “Favorites” heading, each entry includes the book name, author, and a book cover image, offering a visually appealing and organized layout.Users can take immediate action using two prominent buttons: “Read” to start or resume reading, and “Remove” to delete the book from the favorites list. This section is accessible from the left-hand navigation and provides a convenient way to revisit and manage personally selected titles, streamlining the reading experience.

15.png

Figure 4.17: Favorites

4.7.16 Continue Reading

At the top of the Continue Reading section, users are greeted with a warm “Welcome!” message followed by a subtle prompt—“What will you read today?”—designed to encourage continued engagement with their current books. Just below, a prominent search bar allows users to quickly locate books by typing in keywords like titles, authors, or genres, making navigation efficient and user-friendly.This section displays each in-progress book with key information: the book name, author, and cover image, along with a progress bar visually showing the reading percentage .To the right of each entry, users can choose to “Resume” reading instantly or “Remove” the book from the list, enabling seamless control over their reading journey.

16.png

Figure 4.18: Continue Reading

4.7.17 Profile

The Profile section serves as the central hub for managing personal details and customizing reading preferences. Upon entering, users are greeted with a welcoming "Welcome!" message and a convenient search bar to explore titles, authors, or genres across the platform. The profile area displays key user information, including Name, Email, Date of Birth, Gender, Preferred Genre, and Preferred Author, allowing for a highly tailored experience.Below this information, users can select "Edit Profile" to update their personal details or click on "Reading History" to review previously explored content. This section ensures users stay in control of their reading journey while enjoying a personalized and engaging interface.

17.png

Figure 4.19: Profile

4.7.18 Edit Profile

The Edit Profile section empowers users to update their personal details and refine their reading experience. Displayed under the welcoming “Welcome!” header and supported by a handy search bar, this section provides editable fields for Name, Email, and Date of Birth, along with dropdown menus for selecting Gender, a Preferred Author, and a Preferred Genre.Designed with simplicity and clarity, each input field ensures users can personalize their profile in just a few clicks. Once the desired changes are made, clicking the prominent “Save Changes” button finalizes the updates—ensuring the platform continues to deliver relevant and engaging book recommendations.

18.png

Figure 4.20: Edit Profile

4.7.19 Reading History

The Reading History section gives users a comprehensive overview of the books they've finished. Just below the friendly Welcome greeting and a convenient search bar, users find a clearly labeled Reading history area that showcases each completed title with essential details like the Book Name, Author, and Rating, along with any Feedback provided and the Date Finished.Each book entry includes two interactive options: a blue Re-read button for jumping back into favorite stories and a red Remove from history link to declutter the list. This section offers a streamlined way to reflect on past reads, manage personal book records, and revisit stories that made an impact.

19.png

Figure 4.21: Reading History

4.7.20 Logout Confirmation

The Logout Confirmation popup appears when the user initiates the logout process. It displays a clear message asking if the user is sure they want to log out. Two options are provided: a Cancel button to stay logged in and a red Yes, Logout button to confirm and exit the session. This ensures user intent and prevents accidental logouts.

20.png

Figure 4.22: Logout Confirmation

Chapter 5

TESTING AND IMPLEMENTATION

5.1 Testing

5.1.1 Test Case-1

No	Date	Action	Expected Result	Actual Result	Pass?
1	25/08/25	Login with username and password	User should login	User can login successfully	Yes
2	25/08/25	Registration of user	User should register	User can register successfully	Yes
3	25/08/25	Set personalized favorite genre and author	To get personalized suggestions for books	User got personalized suggestions for books	Yes

Table 5.1: Test Case-1

5.1.2 Test Case-2

No	Date	Action	Expected Result	Actual Result	Pass?
1	10/09/25	Admin manages book data and view user profiles	Admin should manage books and view user profiles	Admin can manages books and view user profiles	Yes
2	10/09/25	Search for books or enter one already liked	User should search for books and view favorites	User can search for books and view favorites	Yes

Table 5.2: Test Case-2

5.1.3 Test Case-3

No	Date	Action	Expected Result	Actual Result	Pass?
1	29/09/25	User view recommended books with short description	User should view recommended books with short description	User can view recommended books with short description	Yes
2	29/09/25	User read,rates and add into favorite list	User should read,rates and add into favorite list	User can read,rates and add into favorite list	Yes
3	29/09/25	User give feedback to recommendations	User should give feedback to recommendations	User can give feedback to recommendations	Yes

Table 5.3: Test Case-3

5.1.4 Test Case-4

No	Date	Action	Expected Result	Actual Result	Pass?
1	13/10/25	Admin review feedback and adjust recommendations	Admin should review feedback and adjust recommendations	Admin can review feedback and adjust recommendations	Yes
2	13/10/25	User view favorite list and reading history	User should view favorite list and reading history	User can view favorite list and reading history	Yes

Table 5.4: Test Case-4

5.2 Implementation

After testing, the next phase for the AI-Based Personalized Book Recommendation Engine is the Implementation phase. This phase involves converting the system design into a fully functional web application that delivers personalized book recommendations to users. The implementation includes configuring user management, book management, recommendation algorithms, and deploying the system for users and administrators.

Identify Deployment Environment: Choose a secure server, either cloud-based or local, supporting stable internet connectivity and safe handling of sensitive data such as user profiles, ratings, and feedback. The system is compatible with Windows 10 or Ubuntu.

Install Required Software: Set up Python and Flask for backend development, along with MySQL for database management. Install frontend libraries using HTML, CSS, and JavaScript. Use VS Code as the development IDE and ensure compatibility with browsers like Chrome and Firefox. Machine learning operations for recommendation are implemented using scikit-learn.

Backend Application Development: Using Flask, implement core functionalities including user registration, login, profiles, preferences, ratings, and feedback. The Book Module stores metadata, summaries, genres, and content necessary for content-based filtering. The Recommendation Engine applies TF-IDF, Cosine Similarity, and Collaborative Filtering to generate personalized suggestions.

Web Application Development: Develop a responsive frontend using HTML, CSS, and JavaScript. Connect frontend features to Flask routes to allow users to search books, view recommended books, rate and review books, mark favorites, and track reading history.

Deployment and Training: Deploy the Flask application on the selected hosting server, ensuring secure access for users and administrators. Provide documentation and guidance for administrators to manage book records and monitor user activity. Offer simple user guides for readers to utilize features like personalized recommendations and reading progress tracking.

Maintenance and Monitoring: Conduct regular updates, bug fixes, and system monitoring. Schedule database backups and monitor server performance to ensure stable and responsive operation. Ensure recommendation algorithms continue to function efficiently and provide accurate results.

Continuous Improvement: Gather feedback from users and administrators to refine system functionality and improve the recommendation engine. Update datasets, optimize algorithms, and enhance the user interface based on interactions to ensure the system remains adaptive, efficient, and user-friendly.

Chapter 6

RESULT AND DISCUSSION

The AI-Based Personalized Book Recommendation Engine effectively delivers accurate, personalized book suggestions based on individual user preferences. By leveraging TF-IDF, Cosine Similarity, and Collaborative Filtering algorithms, the system analyzes user data, book metadata, and reading patterns to generate relevant recommendations. It efficiently manages user registration, profile creation, feedback collection, and dynamic preference updates, allowing continuous adaptation to changing interests. Testing with a structured dataset of book titles, authors, genres, and ratings demonstrated that the system provides highly consistent suggestions, improving in accuracy as users interact more. The admin module performs smoothly, enabling easy book management, data updates, and system monitoring.

The system meets its goal of simplifying book discovery by automating recommendations through AI analysis, saving users time and enhancing reading satisfaction. Combining content-based and collaborative filtering balances personalization with diversity. Challenges such as dataset quality and similarity computation optimization were addressed through preprocessing and algorithm fine-tuning. Overall, the system offers a scalable, user-friendly solution for both readers and administrators. Future improvements may include deep learning models, natural language processing for sentiment analysis, and real-time updates to further enhance accuracy and user experience.

Chapter 7

CONCLUSION

The AI-Based Personalized Book Recommendation Engine is developed to simplify and enhance the process of book discovery for readers by delivering intelligent, accurate, and personalized recommendations. By integrating advanced AI algorithms such as TF-IDF, Cosine Similarity, and Collaborative Filtering, the system effectively understands individual reading preferences and provides meaningful book suggestions that align with user interests. The platform automates several core functions, including user registration, profile creation, book search, feedback collection, and recommendation generation, thereby reducing manual effort and improving the overall user experience. Administrators benefit from efficient management tools that allow them to update book records, monitor user activity, and maintain data accuracy. The system successfully bridges the gap between readers and vast digital libraries by transforming raw data into insightful, user-specific suggestions that evolve over time based on interaction and feedback.

As the system continues to evolve, it is expected to further improve recommendation accuracy, user engagement, and system scalability. By offering personalized results and continuously learning from user behavior, the AI-Based Personalized Book Recommendation Engine contributes significantly to promoting reading habits and providing users with a seamless, engaging experience in discovering new books. Ultimately, the system stands as a powerful step forward in utilizing artificial intelligence for personalization, efficiency, and user satisfaction in digital reading platforms.

7.1 Future Work

Looking ahead, the AI-Based Personalized Book Recommendation Engine aims to incorporate several key enhancements to make the recommendation process even smarter and more interactive. One major development will be the introduction of a mobile application that allows users to conveniently access personalized book suggestions, manage profiles, rate books, and receive instant notifications on new recommendations directly from their smartphones. This mobile integration will improve accessibility and user engagement, making book discovery effortless anytime and anywhere.

Additionally, the future version of the system will explore deep learning techniques and Natural Language Processing (NLP) to better understand book content and user sentiments from reviews, leading to more context-aware and emotionally relevant recommendations. The incorporation of real-time recommendation updates, social features such as friend-based suggestions, and data analytics dashboards for administrators will further enhance system performance and transparency. These advancements will ensure that the system remains adaptive, insightful, and user-centric — continuing to transform how readers explore and connect with books through the power of artificial intelligence.

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APPENDICES

SCREENSHOTS

Landing page

The Landing Page of the project serves as an inviting introduction to Book Bot — an intelligent reading companion designed to help users discover their next great read. It features a modern and visually appealing interface with a bold headline, a brief description, and easy navigation options such as "Start Your Journey" and "Sign In" buttons. A background image of a cozy library enhances the page's aesthetic, creating a warm and book-friendly atmosphere. This page provides users with a clear overview of the platform's purpose and encourages them to begin their personalized reading journey effortlessly..

S1.png

Landing page

Registration page

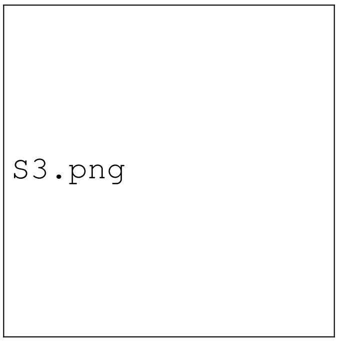
The Registration Page of the project allows new users to create an account and join Book Bot by providing essential personal details. Users are required to enter their Full Name, Email Address, Password, Phone Number, Date of Birth, and Gender. The interface features a clean and modern design with a gradient background and an intuitive form layout. Password strength validation is included to ensure account security, and passwords are securely hashed before storage. The left panel welcomes users with a short message — “Join Book Bot” — encouraging them to begin their personalized reading journey. This page ensures a smooth and secure onboarding process for every new user..

S2.png

Registration page

Login page

The Login Page of the project provides users with a secure and convenient way to access their Book Bot account. It features a simple and user-friendly interface where users can sign in using their Email Address and Password. The design includes a modern gradient background on the left with a welcoming message — “Welcome Back!” — encouraging users to continue their reading journey. Passwords are verified securely to ensure account safety, and a success message confirms when registration has been completed. Additionally, a quick link is provided for new users to create an account if they haven’t registered yet. This page ensures smooth and secure access to the platform’s personalized reading features.



Login page

Preference selection

The Preference Selection Page allows users to personalize their reading experience by selecting their favorite author. This customization enables the Book Bot system to generate tailored book recommendations based on the user's literary preferences. The interface is intuitive and visually engaging, featuring a dark-themed layout with vibrant highlight effects and author selection cards, each represented with an avatar and the author's name. A search bar is provided for quickly locating a preferred author from the list. Upon logging in, the user is greeted with a personalized welcome message along with a prompt encouraging them to set their author and genre preferences to continue. Only one author can be selected at a time, ensuring focused recommendations. The clean design and interactive elements make the setup process simple and user-friendly

S4.png

Preference selection

User dashboard

The User Dashboard Page of Book Bot is a personalized hub designed to enhance each reader's discovery experience. Upon logging in, users are greeted with a warm welcome message and a sleek, modern interface featuring a search bar for quick access to books, authors, and genres. The left-hand sidebar provides smooth navigation through key sections such as Recommended For You, Explore by Category, Recently Rated, Favorites, and Continue Reading—enabling users to seamlessly resume or discover new reads based on their interests. Each section is thoughtfully organized to reflect user preferences set during onboarding. With vibrant accents on a dark-themed layout, interactive cards, and intuitive design, the dashboard offers a user-friendly and immersive platform for discovering tailored book recommendations and managing personal reading activity. Profile and logout options are conveniently located for account management.

S5.png

User dashboard

Recommended books

The Recommended Books Page provides users with personalized book suggestions based on their selected preferences, such as favorite author and genre. At the top, users can see which preferences are being applied—for example, Aaron Jefferson — Thriller. A dynamic search bar allows readers to explore further by title, author, genre, or ISBN. This page highlights the Top 5 Trending Books, showcasing the most highly rated titles across different genres, complete with ratings, genre tags, and author names. Each book card features interactive buttons to instantly Add to Favorites or Read Now, making it easy to engage with content. The clean, dark-themed UI with vibrant call-to-action buttons ensures a smooth and visually appealing experience as users explore books tailored to their tastes..

S6.png

Recommended books

Book description

The Book Description Page provides readers with comprehensive information about a selected title, allowing for an immersive and personalized browsing experience. At the top, users can view the book's title, author, genre, and average rating, giving a quick overview of its key details. The central section features a brief yet engaging description that introduces the book's main themes and tone, helping readers decide if it matches their interests. Interactive buttons such as Add to Favorites and Read Now enable users to easily save or start reading the book directly. A consistent dark-themed interface with vibrant purple accents enhances readability and creates a modern, elegant aesthetic. The sidebar navigation allows seamless movement across other sections—such as Home, Recommended For You, and Favorites—making the Book Details Page both visually appealing and highly functional.

S7.png

Book description

Reading interface

The Reading Interface Page provides a focused and interactive environment for reading within the Book Bot application. Users can highlight important text, save their progress, and adjust font size to enhance comfort and accessibility. The Save Progress feature ensures readers can easily continue from where they stopped, while the page tracker displays their current position in the book. Quick-access buttons like Continue Reading and Exit Highlight make navigation smooth and efficient. With its clean, dark-themed layout and clear typography, the page minimizes distractions and promotes an enjoyable, immersive reading experience.

S8.png

Reading interface

Rating and Feedback

The Rating and Feedback Page allows users to share their opinions and experiences after reading a book. It features a star-based rating system where readers can rate the book from one to five stars, with descriptive feedback labels such as "Very Good." Below the rating section, users can leave written feedback to express their thoughts, highlight favorite aspects, or suggest improvements. Once complete, the Submit button records the user's response, contributing to the overall book rating and helping future readers make informed choices. The page maintains a clean, dark-themed layout consistent with the rest of the Book Bot interface, ensuring clarity, ease of use, and a pleasant user experience.

S9.png

Rating and Feedback

Recently rated

The Recently Rated Books Page displays all the books a user has recently reviewed and rated within the Book Bot application. It shows each book's title, author, rating, feedback, and the date and time the rating was submitted. This allows users to easily track their past interactions and revisit books they've enjoyed or critiqued. A summary section at the top indicates the total number of ratings, providing a quick overview of reading activity. The consistent dark-themed interface and neatly organized book cards ensure clarity, readability, and a smooth browsing experience.

S10.png

Recently rated

Favorite books

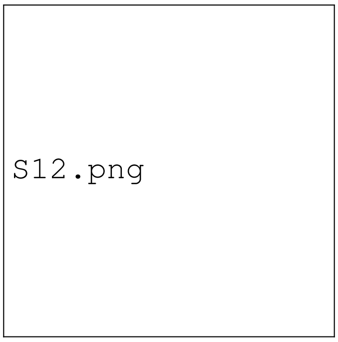
The Favorite Books Page displays a user's personal collection of saved or favorited books within the Book Bot application. It includes a search bar that allows users to quickly find specific books by title, author, genre, or ISBN. Each book card shows the title, author, and average rating, along with two interactive buttons — Read to continue reading instantly and Remove to delete the book from the favorites list. This page helps users easily access their preferred reads in one place. Its clean, dark-themed interface with clear typography and color-coded buttons ensures both functionality and visual appeal.

S11.png

Favorite books

Continue reading

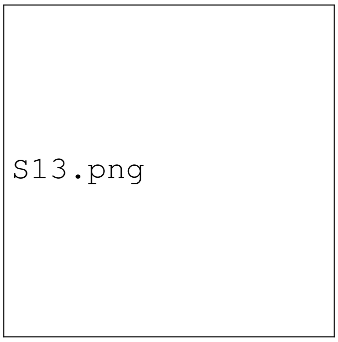
The Continue Reading Page allows users to easily pick up where they left off in their books. Each book card displays the title, author, and reading progress represented by a progress bar, making it simple to track how much of the book has been completed. Users can choose to resume reading directly or remove a title from the list. This feature helps maintain reading continuity and organization, ensuring a seamless experience for ongoing reads. The page's dark-themed interface, paired with clear progress indicators and colorful action buttons, provides both clarity and convenience for regular readers.



Continue reading

Reading history

The Reading History Page displays all the books a user has completed reading in the Book Bot application. Each entry includes the book title, author, genre, user rating, feedback, and the completion date, allowing users to reflect on their reading journey. Interactive buttons such as Re-read and Remove from History give users control over managing their past reads. This page helps track reading achievements and easily revisit favorite books. The sleek, dark-themed interface with structured layouts and bright action buttons ensures an organized and visually engaging user experience.



Reading history

Admin dashboard

The Admin Dashboard serves as the central control panel for managing the Book Bot system. It provides administrators with essential tools to oversee user activities, manage book data, and monitor system performance. The interface greets the admin with a personalized welcome message and a search bar for quickly locating books by title, author, genre, or ISBN. Key sections include View User Profile for analyzing user behavior and preferences, and Manage Books, which allows adding, editing, and organizing the catalog efficiently. Additional options like View Ratings and View Feedback help admins gain insights into user engagement and satisfaction. The dashboard's sleek dark-themed layout, intuitive navigation, and well-structured design ensure an efficient and visually appealing administrative experience.

S14.png

Admin dashboard

View user profile

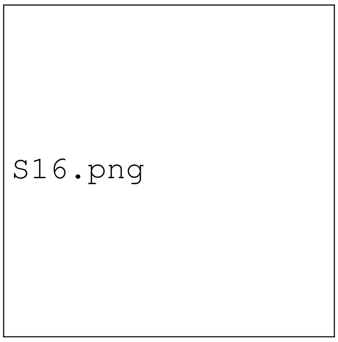
The View User Profile page in the admin panel displays a detailed list of all registered users on the Book Bot platform. It includes essential information such as User ID, Username, Email, Phone Number, Date of Birth, Gender, Preferred Author, and Preferred Genre. Each entry has a “View Details” option, allowing the admin to access more in-depth data about individual users, such as their reading activities, preferences, and feedback. This page helps administrators analyze user trends, monitor engagement, and ensure a personalized reading experience. Its organized table layout and clean dark interface make it easy to navigate and manage user information efficiently.

S15.png

View user profile

View books

The View Books page in the admin panel serves as the Book Catalog of the Book Bot system. It allows administrators to browse, search, and manage all available books efficiently. Each book entry displays key information such as Title, Author, ISBN, and Genre, along with an Action button to view more details. The page also includes a search bar, enabling quick access to specific books by title or ID. The interface is designed with a clean, dark theme and provides an overview of the total number of books in the system, ensuring smooth catalog management and easy navigation for admins.



View books

Add books

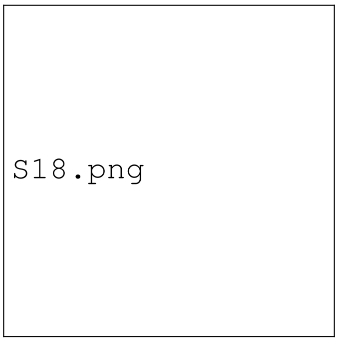
The Add Books page enables administrators to efficiently expand the Book Bot catalog by adding new titles to the system. Admins can input key details such as Book Title, Author, ISBN, and Genre, with real-time validation ensuring that the ISBN meets the 13-digit requirement. A Book Preview section provides a live display of the entered details, allowing admins to verify information before submission. The interface includes convenient navigation buttons to Add Book, View Books, or return to the Home page. Designed with a modern dark-themed layout, this page ensures a smooth and accurate process for maintaining the digital library's growing collection.

S17.png

Add books

View ratings

The View Ratings page allows administrators to monitor and analyze how users rate different books within the Book Bot system. It features a search bar that enables filtering by book name, ISBN, or ID, helping admins quickly locate specific titles. The page displays a visual rating distribution chart, giving a clear overview of how readers have rated a selected book. Essential book details—such as Title, Author, Genre, and ISBN—are shown alongside the graph. With its clean dark interface and visual analytics, this page helps admins track user engagement and identify books that perform well or need attention based on user feedback trends.



View ratings

View feedback

T he View Feedback page enables administrators to review user opinions and comments about books in the Book Bot system. It includes a search bar for locating specific books using the title, ISBN, or ID. Once a book is selected, detailed feedback entries are displayed, showing each user's ID, comment, and the exact date and time it was submitted. This feature helps admins understand reader satisfaction, identify popular books, and gather insights for improving recommendations. The page's dark, modern interface makes reading and managing user feedback clear and visually organized.

S19.png

View feedback